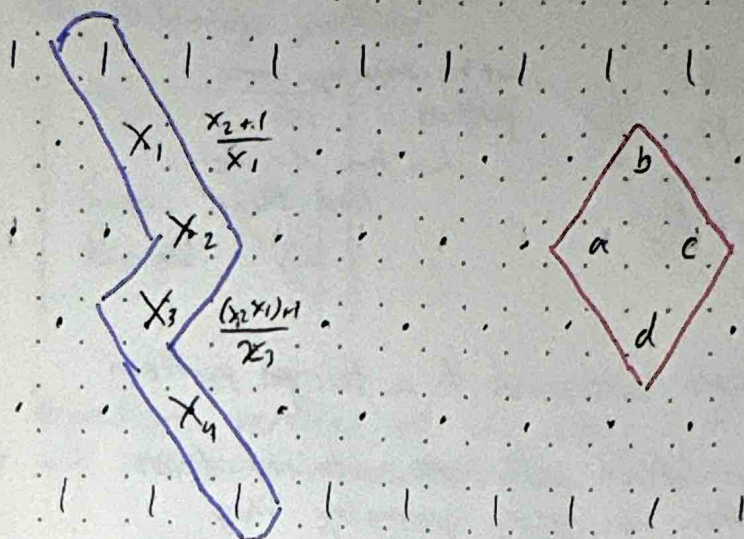


(Recall) Frieze Patterns: (cont)

EX: consider $n=7$



→ the entries are positive integers/real or just variables

↳ $n-1$ rows

↳ top/bottom rows are all 1's

↳ $\overline{ad-bc} = \underline{1}$

How can we construct frieze patterns — the space of all frieze patterns

↳ use "lattice paths" (frieze pattern is our "lattice")

→ all entries are expressed in terms of x_i for $i=1, \dots, n-3$ as subtraction-free rational expressions.

→ space of real frieze patterns $\cong (\mathbb{R}_{>0})^{n-3}$

We see an instance of Laurent phenomenon, where these rational expressions are positive-integer Laurent polynomials.

We have the following non-trivial fact:

→ any lattice path gives an integer frieze pattern by specializing $x_1 = x_2 = \dots = x_{n-3} = 1$.

↳ there are $\frac{2^{n-4} \cdot n}{2}$ many of them.

n^{th} Catalan number

n	4	5	6
$2^{n-4} \cdot n$	2	5	12
C_n	2	5	14

we're missing some patterns...

↳ how do we find them?

Note: every lattice path corresponds to a frieze pattern

↳ if two lattice paths are obtained from each other via glide symmetry, they correspond to the same frieze pattern.

Consider another view of frieze patterns.

Def'n (frieze pattern):

↳ an array (matrix) $P = (p_{ij})$ for $i \leq j \leq n+1$ and $i, j \in \mathbb{Z}$ such that for any 2×2 minor, we have

$$\begin{vmatrix} p_{ij} & p_{i,j+1} \\ p_{i+1,j} & p_{i+1,j+1} \end{vmatrix} = 1$$

$$\text{and } p_{i, i+1} = p_{i, i+n+1} = 1.$$

We can extend this to the $\infty \times \infty$ matrix $P = (p_{ij})$, for $i, j \in \mathbb{Z}$ such that the following hold

1) $p_{ij} = -p_{i, j+n}$

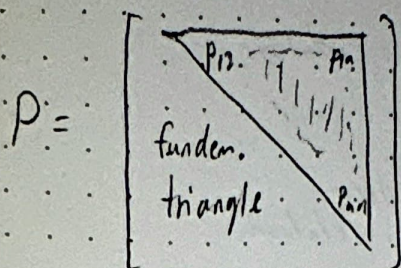
2) $p_{ii} = 0$

by n -periodicity: $p_{ij} = p_{i+n, j+n} \Leftrightarrow p_{ij} = -p_{i+n, j}$

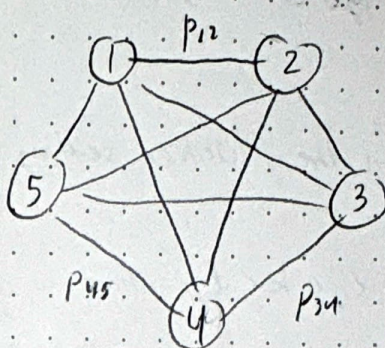
by glide symmetry, we have $p_{ij} = p_{j,i+n} \Leftrightarrow p_{ij} = -p_{ji}$

→ since we have C_{n-2} many frieze patterns of order n , it makes sense to consider a bijection with triangulations of n -gons.

Notice the following pattern:

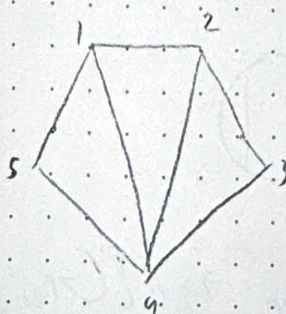


For our bijection, vertices of our n -gon will be labeled by $[n]_n$ and edges will correspond to entries p_{ij} from our frieze pattern.



all possible edges - not a triangulation

EX:



$$p_{12} = p_{23} = p_{45} = p_{15} = 1$$

$$p_{14} = p_{24} = 1$$

zig-zag

→ some triangulations correspond to lattice paths

↳ those correspond to frieze patterns obtainable from lattice paths

→ triangulations not corresponding to lattice paths get the remaining patterns

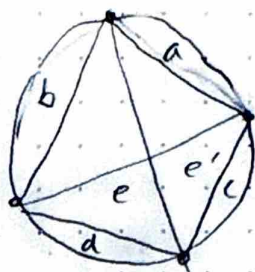


→

We can now notice a connection to (cluster Algebras), but first, we revisit work of the ancient greeks.

Thm (Ptolemy):

↳



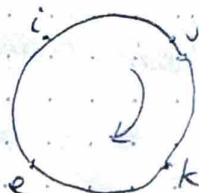
$$e \cdot e' = ad + bc$$

↑
three term
Plücker relation!

Def'n (Ptolemy Algebra):

↳ P_n (over some field $\mathbb{F}$) generated by $\binom{n}{2}$ variables Δ_{ij} for $i < j \leq n$.

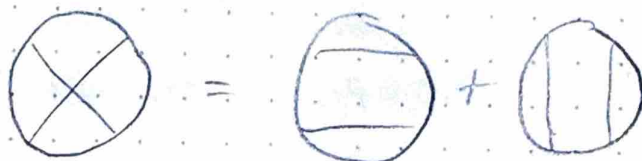
Lets further consider things in the Plücker setting.



for $i < j < k < l$, where

$$\Delta_{ik} \cdot \Delta_{je} = \Delta_{ij} \cdot \Delta_{ke} + \Delta_{ic} \cdot \Delta_{jk}$$

notice that, we can equiv express

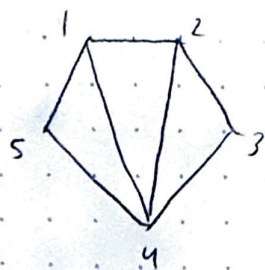


Basically, $P_n \cong$ Coordinate ring of $Gr_{2n} \rightarrow \mathbb{F}[Gr_{2n}]$.

Recall that $\dim(Gr_{2n}) = n(n-2)$. Hopefully there is a choice of $2(n-2)+1$ algebraically independent Plücker variables Δ_{ij} .

↳ any triangulation of an n -gon gives such a subset of Δ_{ij} 's.

52.



$$\rightarrow \Delta_{12}, \Delta_{23}, \Delta_{34}, \Delta_{45}, \Delta_{15}, \Delta_{14}, \Delta_{24}$$

All other Δ_{ij} 's are expressed in terms of this "cluster," for instance

$$\Delta_{13} = \frac{\Delta_{12} \cdot \Delta_{34} + \Delta_{14} \cdot \Delta_{23}}{\Delta_{24}}$$